

**Vishwakarma Institute of Technology, Pune**

**Practical Implementation Sheet**

Department: CSE-AI      Semester: 5      Academic Year: 2026-27

Division: E      Batch: 2      Practical No: 6

Roll no: 41      PRN No: 12410751      Name of Student: Rucha Narwade

**Subject Name : Deep Learning**

**Problem Statement**

Design and implement a Convolutional Neural Network (CNN) for image classification using the Tomato or Soybean disease dataset.

**Algorithm**

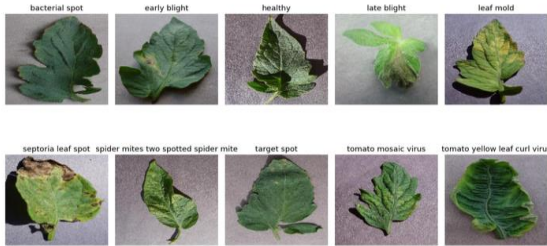
1. Begin
2. Collect the Tomato Disease image dataset containing healthy and infected tomato leaves.
3. Determine the different categories available in the dataset.
4. Convert all input images to a fixed size of  $128 \times 128$  pixels.
5. Scale the pixel values between 0 and 1 by dividing them by 255.
6. Separate the images into training and validation data.
7. Initialize the Convolutional Neural Network.
8. Apply the first convolution operation using 32 filters of size  $3 \times 3$ .
9. Use the ReLU activation function to introduce non-linearity.
10. Perform  $2 \times 2$  Max Pooling to reduce the feature-map dimensions.
11. Apply a second convolution layer containing 64 filters.
12. Apply ReLU activation and  $2 \times 2$  Max Pooling.
13. Add a third convolution layer with 128 filters.
14. Apply ReLU activation followed by another Max Pooling operation.
15. Flatten the resulting feature maps into a single vector.
16. Connect the flattened features to a fully connected layer containing 128 neurons.
17. Apply a Dropout rate of 0.5 to minimize overfitting.
18. Add the final classification layer with Softmax activation.
19. Configure the network using the Adam optimizer and categorical cross-entropy loss.
20. Train the network with the prepared training images for the selected number of epochs.
21. Check the model performance on the validation images after training.
22. Measure the performance using accuracy, loss, precision, recall, F1-score, and a confusion matrix.
23. Provide a new tomato leaf image as input to the trained network.
24. Prepare the new image by resizing and normalizing it.

25. Feed the processed image into the trained CNN.
26. Select the disease category having the highest prediction probability.
27. Show the detected disease along with its confidence percentage.
28. End

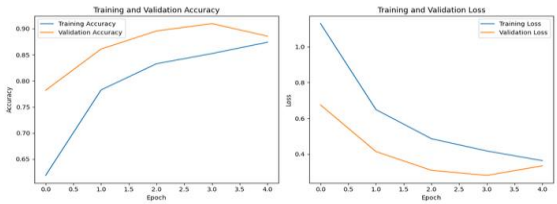
OUTPUT :

## CNN IMAGE CLASSIFICATION - OUTPUT

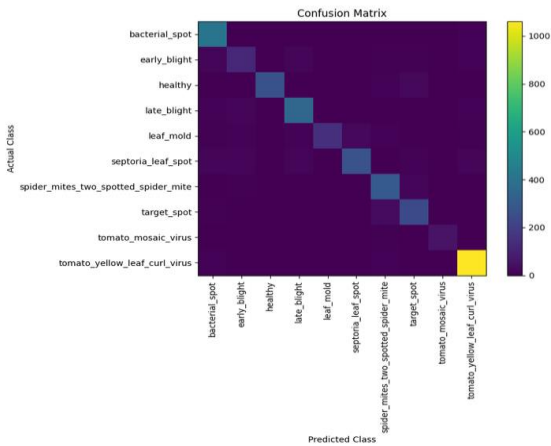
### Dataset Samples



### Accuracy and Loss



### Confusion Matrix



### Model Evaluation and Prediction

Validation Accuracy : 0.8857378959655762%

Validation Loss : 0.33496996760368347

Predicted Disease : spider mites two spotted spider mite

Confidence : 56.10%

### Classification Report

| precision    | recall                        | f1-score | support |      |      |     |
|--------------|-------------------------------|----------|---------|------|------|-----|
|              | bacterial_spot                | 0.86     | 0.97    | 0.91 | 428  |     |
|              | early_blight                  | 0.68     | 0.66    | 0.67 | 181  |     |
|              | healthy                       | 0.99     | 0.89    | 0.94 | 298  |     |
|              | late_blight                   | 0.88     | 0.88    | 0.88 | 397  |     |
|              | leaf_mold                     | 0.96     | 0.68    | 0.80 | 208  |     |
|              | septoria_leaf_spot            | 0.91     | 0.77    | 0.83 | 354  |     |
| spider_mites | two_spotted_spider_mite       | 0.79     | 0.79    | 0.91 | 0.85 | 327 |
|              | target_spot                   | 0.79     | 0.83    | 0.81 | 289  |     |
|              | tomato_mosaic_virus           | 0.89     | 0.84    | 0.86 | 67   |     |
|              | tomato_yellow_leaf_curl_virus | 0.95     | 0.98    | 0.96 | 1083 |     |
|              | accuracy                      |          | 0.89    | 3632 |      |     |
|              | macro avg                     | 0.87     | 0.84    | 0.85 | 3632 |     |
|              | weighted avg                  | 0.89     | 0.89    | 0.88 | 3632 |     |